

CMMi Process Areas

For Development

As per CMMi, Process Area is,

"A cluster of related practices in an area that, when implemented collectively, satisfies a set of goals considered important for making improvement in that area."

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1. Project Planning (PP)

A Project Management process area at Maturity Level 2.

The purpose of Project Planning (PP) is to establish and maintain plans that define project activities.

The Project Planning process area involves the following:

- Developing the project plan
- Interacting with stakeholders appropriately
- Getting commitment to the plan
- Maintaining the plan

Specific Practices by Goal

- SG 1 Establish Estimates
 - SP 1.1 Estimate the Scope of the Project
 - SP 1.2 Establish Estimates of Work Product and Task Attributes
 - SP 1.3 Define Project Lifecycle Phases
 - SP 1.4 Estimate Effort and Cost
- SG 2 Develop a Project Plan
 - SP 2.1 Establish the Budget and Schedule
 - SP 2.2 Identify Project Risks
 - SP 2.3 Plan Data Management
 - SP 2.4 Plan the Project's Resources
 - SP 2.5 Plan Needed Knowledge and Skills
 - SP 2.6 Plan Stakeholder Involvement
 - SP 2.7 Establish the Project Plan
- SG 3 Obtain Commitment to the Plan
 - SP 3.1 Review Plans that Affect the Project
 - SP 3.2 Reconcile Work and Resource Levels
 - SP 3.3 Obtain Plan Commitment

2. Project Monitoring and Control (PMC)

A Project Management process area at Maturity Level 2.

The purpose of Project Monitoring and Control (PMC) is to provide an understanding of the project's progress so that appropriate corrective actions can be taken when the project's performance deviates significantly from the plan.

Specific Practices by Goal

- SG 1 Monitor the Project Against the Plan
 - SP 1.1 Monitor Project Planning Parameters
 - SP 1.2 Monitor Commitments
 - SP 1.3 Monitor Project Risks
 - SP 1.4 Monitor Data Management
 - SP 1.5 Monitor Stakeholder Involvement
 - SP 1.6 Conduct Progress Reviews
 - SP 1.7 Conduct Milestone Reviews

- SG 2 Manage Corrective Action to Closure
 - SP 2.1 Analyze Issues
 - SP 2.2 Take Corrective Action
 - SP 2.3 Manage Corrective Actions

3. Process and Product Quality Assurance (PPQA)

A Support process area at Maturity Level 2.

The purpose of Process and Product Quality Assurance (PPQA) is to provide staff and management with objective insight into processes and associated work products.

PPQA process area involves the following:

- Objectively evaluating performed processes, work products, and services against the applicable process descriptions, standards, and procedures
- Identifying and documenting noncompliance issues
- Providing feedback to project staff and managers on the results of quality assurance activities
- Ensuring that noncompliance issues are addressed

Specific Practices by Goal

- SG 1 Objectively Evaluate Processes and Work Products
 - SP 1.1 Objectively Evaluate Processes
 - SP 1.2 Objectively Evaluate Work Products
- SG 2 Provide Objective Insight
 - SP 2.1 Communicate and Resolve Noncompliance Issues
 - SP 2.2 Establish Records

4. Requirements Management (REQM)

A Project Management process area at Maturity Level 2.

The purpose of Requirements Management (REQM) is to manage requirements of the project's products and product components and to ensure alignment between those requirements and the project's plans and work products.

Specific Practices by Goal

- SG 1 Manage Requirements
 - SP 1.1 Understand Requirements
 - SP 1.2 Obtain Commitment to Requirements
 - SP 1.3 Manage Requirements Changes
 - SP 1.4 Maintain Bidirectional Traceability of Requirements
 - SP 1.5 Ensure Alignment Between Project Work and Requirements

5. Supplier Agreement Management (SAM)

A Project Management process area at Maturity Level 2.

The purpose of Supplier Agreement Management (SAM) is to manage the acquisition of products from suppliers.

Specific Practices by Goal

- SG 1 Establish Supplier Agreements
 - SP 1.1 Determine Acquisition Type
 - SP 1.2 Select Suppliers
 - SP 1.3 Establish Supplier Agreements
- SG 2 Satisfy Supplier Agreements
 - SP 2.1 Execute the Supplier Agreement
 - SP 2.2 Accept the Acquired Product
 - SP 2.3 Ensure Transition of Products

6. Configuration Management (CM)

A Support process area at Maturity Level 2.

The purpose of Configuration Management (CM) is to establish and maintain the integrity of work products using configuration identification, configuration control, configuration status accounting, and configuration audits.

Specific Practices by Goal

- SG 1 Establish Baselines
 - SP 1.1 Identify Configuration Items
 - SP 1.2 Establish a Configuration Management System
 - SP 1.3 Create or Release Baselines
- SG 2 Track and Control Changes
 - SP 2.1 Track Change Requests
 - SP 2.2 Control Configuration Items
- SG 3 Establish Integrity
 - SP 3.1 Establish Configuration Management Records
 - SP 3.2 Perform Configuration Audits

7. Measurement and Analysis (MA)

A Support process area at Maturity Level 2.

The purpose of Measurement and Analysis (MA) is to develop and sustain a measurement capability used to support management information needs.

Specific Practices by Goal

- SG 1 Align Measurement and Analysis Activities
 - SP 1.1 Establish Measurement Objectives
 - Resources, People, Facilities and Techniques.
 - SP 1.2 Specify Measures

- Information Needs Document, Guidance, Reference and Reporting.
- SP 1.3 Specify Data Collection and Storage Procedures
 - Sources, Methods, Frequency and Owners.
- SP 1.4 Specify Analysis Procedures
 - Rules, Alarms, SPC and Variance.
- SG 2 Provide Measurement Results
 - SP 2.1 Obtain Measurement Data
 - Actual, Plan, Automatic and Manual.
 - SP 2.2 Analyze Measurement Data
 - Evaluate, Drill Down, RCA.
 - SP 2.3 Store Data and Results
 - Store, Secure, Accessible, History and Evidence.
 - SP 2.4 Communicate Results
 - Information Sharing, Dashboards, Up to Date, Simple and Interpret.

8. Decision Analysis and Resolution (DAR)

A Support process area at Maturity Level 3.

The purpose of Decision Analysis and Resolution (DAR) is to analyze possible decisions using a formal evaluation process that evaluates identified alternatives against established criteria.

Specific Practices by Goal

- SG 1 Evaluate Alternatives
 - SP 1.1 Establish Guidelines for Decision Analysis
 - SP 1.2 Establish Evaluation Criteria
 - SP 1.3 Identify Alternative Solutions
 - SP 1.4 Select Evaluation Methods
 - SP 1.5 Evaluate Alternative Solutions
 - SP 1.6 Select Solutions

9. Integrated Project Management (IPM)

A Project Management process area at Maturity Level 3.

The purpose of Integrated Project Management (IPM) is to establish and manage the project and the involvement of relevant stakeholders according to an integrated and defined process that is tailored from the organization's set of standard processes.

Specific Practices by Goal

- SG 1 Use the Project's Defined Process
 - SP 1.1 Establish the Project's Defined Process
 - SP 1.2 Use Organizational Process Assets for Planning Project Activities
 - SP 1.3 Establish the Project's Work Environment
 - SP 1.4 Integrate Plans
 - SP 1.5 Manage the Project Using the Integrated Plans
 - SP 1.6 Establish Teams
 - SP 1.7 Contribute to Organizational Process Assets

- SG 2 Coordinate and Collaborate with Relevant Stakeholders
 - SP 2.1 Manage Stakeholder Involvement
 - SP 2.2 Manage Dependencies
 - SP 2.3 Resolve Coordination Issues

10. Organizational Process Definition (OPD)

A Process Management process area at Maturity Level 3.

The purpose of Organizational Process Definition (OPD) is to establish and maintain a usable set of organizational process assets, work environment standards, and rules and guidelines for teams.

Specific Practices by Goal

- SG 1 Establish Organizational Process Assets
 - SP 1.1 Establish Standard Processes
 - SP 1.2 Establish Lifecycle Model Descriptions
 - SP 1.3 Establish Tailoring Criteria and Guidelines
 - SP 1.4 Establish the Organization's Measurement Repository
 - SP 1.5 Establish the Organization's Process Asset Library
 - SP 1.6 Establish Work Environment Standards
 - SP 1.7 Establish Rules and Guidelines for Teams

11. Organizational Process Focus (OPF)

A Process Management process area at Maturity Level 3

The purpose of Organizational Process Focus (OPF) is to plan, implement, and deploy organizational process improvements based on a thorough understanding of current strengths and weaknesses of the organization's processes and process assets.

Specific Practices by Goal

- SG 1 Determine Process Improvement Opportunities
 - SP 1.1 Establish Organizational Process Needs
 - SP 1.2 Appraise the Organization's Processes
 - SP 1.3 Identify the Organization's Process Improvements
- SG 2 Plan and Implement Process Improvements
 - SP 2.1 Establish Process Action Plans
 - SP 2.2 Implement Process Action Plans
- SG 3 Deploy Organizational Process Assets and Incorporate Experiences
 - SP 3.1 Deploy Organizational Process Assets
 - SP 3.2 Deploy Standard Processes
 - SP 3.3 Monitor the Implementation
 - SP 3.4 Incorporate Experiences into Organizational Process Assets

12. Organizational Training (OT)

A Process Management process area at Maturity Level 3.

The purpose of Organizational Training (OT) is to develop skills and knowledge of people so they can perform their roles effectively and efficiently.

Specific Practices by Goal

- SG 1 Establish an Organizational Training Capability
 - SP 1.1 Establish Strategic Training Needs
 - SP 1.2 Determine Which Training Needs Are the Responsibility of the Organization
 - SP 1.3 Establish an Organizational Training Tactical Plan
 - SP 1.4 Establish a Training Capability
- SG 2 Provide Training
 - SP 2.1 Deliver Training
 - SP 2.2 Establish Training Records
 - SP 2.3 Assess Training Effectiveness

13. Product Integration (PI)

An Engineering process area at Maturity Level 3.

The purpose of Product Integration (PI) is to assemble the product from the product components, ensure that the product, as integrated, behaves properly (i.e., possesses the required functionality and quality attributes), and deliver the product.

Specific Practices by Goal

- SG 1 Prepare for Product Integration
 - SP 1.1 Establish an Integration Strategy
 - SP 1.2 Establish the Product Integration Environment
 - SP 1.3 Establish Product Integration Procedures and Criteria
- SG 2 Ensure Interface Compatibility
 - SP 2.1 Review Interface Descriptions for Completeness
 - SP 2.2 Manage Interfaces
- SG 3 Assemble Product Components and Deliver the Product
 - SP 3.1 Confirm Readiness of Product Components for Integration
 - SP 3.2 Assemble Product Components
 - SP 3.3 Evaluate Assembled Product Components
 - SP 3.4 Package and Deliver the Product or Product Component

14. Requirements Development (RD)

An Engineering process area at Maturity Level 3.

The purpose of Requirements Development (RD) is to elicit, analyze, and establish customer, product, and product component requirements.

Specific Practices by Goal

- SG 1 Develop Customer Requirements
 - SP 1.1 Elicit Needs
 - SP 1.2 Transform Stakeholder Needs into Customer Requirements
- SG 2 Develop Product Requirements
 - SP 2.1 Establish Product and Product Component Requirements
 - SP 2.2 Allocate Product Component Requirements
 - SP 2.3 Identify Interface Requirements
- SG 3 Analyze and Validate Requirements
 - SP 3.1 Establish Operational Concepts and Scenarios
 - SP 3.2 Establish a Definition of Required Functionality and Quality Attributes
 - SP 3.3 Analyze Requirements
 - SP 3.4 Analyze Requirements to Achieve Balance
 - SP 3.5 Validate Requirements

15. Risk Management (RSKM)

A Project Management process area at Maturity Level 3.

The purpose of Risk Management (RSKM) is to identify potential problems before they occur so that risk handling activities can be planned and invoked as needed across the life of the product or project to mitigate adverse impacts on achieving objectives.

Specific Practices by Goal

- SG 1 Prepare for Risk Management
 - SP 1.1 Determine Risk Sources and Categories
 - SP 1.2 Define Risk Parameters
 - SP 1.3 Establish a Risk Management Strategy
- SG 2 Identify and Analyze Risks
 - SP 2.1 Identify Risks
 - SP 2.2 Evaluate, Categorize, and Prioritize Risks
- SG 3 Mitigate Risks
 - SP 3.1 Develop Risk Mitigation Plans
 - SP 3.2 Implement Risk Mitigation Plans

16. Technical Solution (TS)

An Engineering process area at Maturity Level 3

The purpose of Technical Solution (TS) is to select design and implement solutions to requirements. Solutions, designs, and implementations encompass products, product components, and product related lifecycle processes either singly or in combination as appropriate.

Specific Practices by Goal

- SG 1 Select Product Component Solutions
 - SP 1.1 Develop Alternative Solutions and Selection Criteria
 - SP 1.2 Select Product Component Solutions
- SG 2 Develop the Design
 - SP 2.1 Design the Product or Product Component
 - SP 2.2 Establish a Technical Data Package
 - SP 2.3 Design Interfaces Using Criteria
 - SP 2.4 Perform Make, Buy or Reuse Analyses
- SG 3 Implement the Product Design
 - SP 3.1 Implement the Design
 - SP 3.2 Develop Product Support Documentation

17. Validation (VAL)

An Engineering process area at Maturity Level 3.

The purpose of Validation (VAL) is to demonstrate that a product or product component fulfills its intended use when placed in its intended environment.

Specific Practices by Goal

- SG 1 Prepare for Validation
 - SP 1.1 Select Products for Validation
 - SP 1.2 Establish the Validation Environment
 - SP 1.3 Establish Validation Procedures and Criteria
- SG 2 Validate Product or Product Components
 - SP 2.1 Perform Validation
 - SP 2.2 Analyze Validation Results

18. Verification (VER)

An Engineering process area at Maturity Level 3.

The purpose of Verification (VER) is to ensure that selected work products meet their specified requirements.

Specific Practices by Goal

- SG 1 Prepare for Verification
 - SP 1.1 Select Work Products for Verification
 - SP 1.2 Establish the Verification Environment
 - SP 1.3 Establish Verification Procedures and Criteria
- SG 2 Perform Peer Reviews
 - SP 2.1 Prepare for Peer Reviews
 - SP 2.2 Conduct Peer Reviews
 - SP 2.3 Analyze Peer Review Data
- SG 3 Verify Selected Work Products
 - SP 3.1 Perform Verification

- SP 3.2 Analyze Verification Results

19. Organizational Process Performance (OPP)

A Process Management process area at Maturity Level 4.

The purpose of Organizational Process Performance (OPP) is to establish and maintain a quantitative understanding of the performance of selected processes in the organization's set of standard processes in support of achieving quality and process performance objectives, and to provide process performance data, baselines, and models to quantitatively manage the organization's projects.

Specific Practices by Goal

- SG 1 Establish Performance Baselines and Models
 - SP 1.1 Establish Quality and Process Performance Objectives
 - SP 1.2 Select Processes
 - SP 1.3 Establish Process Performance Measures
 - SP 1.4 Analyze Process Performance and Establish Process Performance Baselines
 - SP 1.5 Establish Process Performance Models

20. Quantitative Project Management (QPM)

A Project Management process area at Maturity Level 4.

The purpose of the Quantitative Project Management (QPM) process area is to quantitatively manage the project to achieve the project's established quality and process performance objectives.

Specific Practices by Goal

- SG 1 Prepare for Quantitative Management
 - SP 1.1 Establish the Project's Objectives
 - SP 1.2 Compose the Defined Processes
 - SP 1.3 Select Subprocesses and Attributes
 - SP 1.4 Select Measures and Analytic Techniques
- SG 2 Quantitatively Manage the Project
 - SP 2.1 Monitor the Performance of Selected Sub-processes
 - SP 2.2 Manage Project Performance
 - SP 2.3 Perform Root Cause Analysis

21. Causal Analysis and Resolution (CAR)

A Support process area at Maturity Level 5.

The purpose of Causal Analysis and Resolution (CAR) is to identify causes of selected outcomes and take action to improve process performance.

Specific Practices by Goal

- SG 1 Determine Causes of Selected Outcomes

- SP 1.1 Select Outcomes for Analysis
- SP 1.2 Analyze Causes
- SG 2 Address Causes of Selected Outcomes
 - SP 2.1 Implement Action Proposals
 - SP 2.2 Evaluate the Effect of Implemented Actions
 - SP 2.3 Record Causal Analysis Data

22. Organizational Performance Management (OPM)

A Process Management process area at Maturity Level 5.

The purpose of Organizational Performance Management (OPM) is to proactively manage the organization's performance to meet its business objectives.

Specific Practices by Goal

- SG 1 Manage Business Performance
 - SP 1.1 Maintain Business Objectives
 - SP 1.2 Analyze Process Performance Data
 - SP 1.3 Identify Potential Areas for Improvement
- SG 2 Select Improvements
 - SP 2.1 Elicit Suggested Improvements
 - SP 2.2 Analyze Suggested Improvements
 - SP 2.3 Validate Improvements
 - SP 2.4 Select and Implement Improvements for Deployment
- SG 3 Deploy Improvements
 - SP 3.1 Plan the Deployment
 - SP 3.2 Manage the Deployment
 - SP 3.3 Evaluate Improvement Effects

CMMi Process Area and maturity level Grid

Process Area	Abbreviation	Category	Maturity Level
Project Planning	PP	Project Management	2
Project Monitoring and Control	PMC	Project Management	2
Requirements Management	REQM	Project Management	2
Supplier Agreement Management	SAM	Project Management	2
Configuration Management	CM	Support	2
Measurement and Analysis	MA	Support	2
Process and Product Quality Assurance	PPQA	Support	2
Requirements Development	RD	Engineering	3
Technical Solution	TS	Engineering	3
Product Integration	PI	Engineering	3
Validation	VAL	Engineering	3
Verification	VER	Engineering	3
Organizational Process Definition	OPD	Process Management	3
Organizational Process Focus	OPF	Process Management	3
Organizational Training	OT	Process Management	3
Integrated Project Management	IPM	Project Management	3
Risk Management	RSKM	Project Management	3
Decision Analysis and Resolution	DAR	Support	3
Organizational Process Performance	OPP	Process Management	4
Quantitative Project Management	QPM	Project Management	4
Organizational Performance Management	OPM	Process Management	5
Causal Analysis and Resolution	CAR	Support	5

CMMi Process Area Mapping with QMS Procedures

S.No.	CMMi Process Area	Abbreviation	QMS Procedure Name
1	Project Planning	PP	- Project Management Procedure - Estimation Procedure
2	Project Monitoring and Control	PMC	Project Management Procedure
3	Requirements Management	REQM	Requirement Development Procedure
4	Supplier Agreement Management	SAM	Subcontract Management Procedure (-N.A.-)
5	Configuration Management	CM	Configuration and Change Management Procedure
6	Measurement and Analysis	MA	Metrics Procedure
7	Process and Product Quality Assurance	PPQA	Internal Audit Procedure
8	Requirements Development	RD	Requirement Development Procedure
9	Technical Solution	TS	- Design Procedure - Coding and Unit Testing Procedure
10	Product Integration	PI	- Design Procedure - Testing Procedure - Build and Release Procedure
11	Validation	VAL	- Testing Procedure - Acceptance Testing Procedure
12	Verification	VER	Review Procedure
13	Organizational Process Definition	OPD	Process Improvement Procedure
14	Organizational Process Focus	OPF	Process Improvement Procedure
15	Organizational Training	OT	Training Procedure
16	Integrated Project Management	IPM	Project Management Procedure
17	Risk Management	RSKM	Risk Management Procedure
18	Decision Analysis and Resolution	DAR	Formal Evaluation Procedure
19	Organizational Process Performance	OPP	Organizational Process Performance Procedure
20	Quantitative Project Management	QPM	Quantitative Project Management Procedure
21	Organizational Performance Management	OPM	Organizational Performance Management Procedure
22	Causal Analysis and Resolution	CAR	Causal Analysis and Resolution Procedure